

Part A: Control Systems – PID Controllers (25 Q&A)

1. What does PID stand for?

→ Proportional–Integral–Derivative.

2. What is the main objective of a PID controller?

→ To minimize the error between a system's desired setpoint and its actual output by adjusting control inputs.

3. What is the proportional term responsible for?

→ It produces an output proportional to the current error, improving response speed.

4. What is the integral term responsible for?

→ It eliminates steady-state error by summing past errors over time.

5. What is the derivative term responsible for?

→ It predicts future error by considering the rate of change of the error, improving system stability.

6. Write the general equation for a PID controller.

→ $u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$

7. What is the effect of increasing (K_p)?

→ Faster response but may lead to overshoot and oscillation.

8. What happens if (K_i) is too large?

→ It causes instability and excessive oscillations due to integral windup.

9. What does “integral windup” mean?

→ A condition where the integral term accumulates a large error during saturation, causing overshoot when control resumes.

10. What is the main benefit of the derivative term?

→ It improves transient response and reduces overshoot.

11. Why is derivative action sensitive to noise?

→ Because it amplifies rapid changes in error, including measurement noise.

12. What is a practical way to reduce derivative noise?

→ Use a low-pass filter on the derivative signal.

13. What happens if a controller has only a proportional term?

→ It reduces but does not eliminate steady-state error.

14. What kind of error does the integral term remove?

→ Steady-state (offset) error.

15. What is the effect of too much derivative gain?

→ The system becomes sluggish and noisy.

16. What is meant by “tuning” a PID controller?

→ Adjusting (K_p), (K_i), and (K_d) values for optimal performance.

17. Name one common PID tuning method.

→ Ziegler–Nichols method.

18. What is the Ziegler–Nichols closed-loop tuning rule based on?

→ Ultimate gain and ultimate period of sustained oscillations.

19. What is a “setpoint”?

→ The desired output value of the system.

20. What is the main advantage of a PID controller over a P or PI controller?

→ It provides better transient and steady-state performance.

21. What are the units of the integral gain (K_i)?

→ 1/time (e.g., s^{-1}).

22. What are the units of the derivative gain (K_d)?

→ Time (e.g., seconds).

23. What is “anti-windup”?

→ A correction technique to prevent integrator accumulation during actuator saturation.

24. In which systems is derivative control often avoided?

→ In systems with high noise or discontinuous signals.

25. What is the main challenge in implementing PID control digitally?

→ Discretization errors and sampling delay.



Part B: Digital Electronics – Flip Flops (25 Q&A)

1. What is a flip-flop?

→ A bistable circuit that stores one bit of data (0 or 1).

2. Name the basic types of flip-flops.

→ SR, JK, D, and T flip-flops.

3. What is the key difference between latch and flip-flop?

→ A latch is level-triggered; a flip-flop is edge-triggered.

4. What are the inputs of an SR flip-flop?

→ Set (S) and Reset (R).

5. What happens when both S and R are 1 in an SR flip-flop?

→ The state is invalid or indeterminate.

6. What does JK flip-flop stand for?

→ Jack Kilby flip-flop, named after its inventor.

7. What is the function of a JK flip-flop?

→ It eliminates the invalid condition of the SR flip-flop; when $J = K = 1$, it toggles.

8. What does the D flip-flop do?

→ It stores the value of the input D at the clock edge.

9. What is another name for a D flip-flop?

→ Data or delay flip-flop.

10. What is the characteristic equation of a D flip-flop?

→ $Q_{next} = D$

11. What is the characteristic equation of a JK flip-flop?

→ $Q_{next} = J\overline{Q} + \overline{K}Q$

12. What is the characteristic equation of a T flip-flop?

→ $Q_{next} = T \oplus Q$

13. What is a clock signal used for in flip-flops?

→ To synchronize data transfer or state change.

14. Define edge triggering.

→ The flip-flop changes state only on a specific clock transition (rising or falling edge).

15. What is meant by setup time?

→ The minimum time before the clock edge that input data must remain stable.

16. What is hold time?

→ The minimum time after the clock edge that input data must remain stable.

17. What is meant by propagation delay?

→ The time taken for a change in input to cause a change in output.

18. What is the toggle condition in a T flip-flop?

→ When $T = 1$, the output toggles on every clock edge.

19. How can a T flip-flop be implemented using a JK flip-flop?

→ By connecting $J = K = T$.

20. How can a D flip-flop be implemented using a JK flip-flop?

→ By connecting $J = D$ and $K = D'$.

21. What is a master-slave flip-flop?

→ A flip-flop made of two stages to avoid race-around problems.

22. What is race-around condition?

→ When multiple toggles occur within one clock pulse due to long propagation delay.

23. How is race-around avoided?

→ By using edge triggering or master-slave configuration.

24. What is a clocked flip-flop?

→ A flip-flop that changes state only when triggered by a clock pulse.

25. Mention one application of flip-flops.

→ Used in counters, registers, and memory circuits.